

Book Review

Natural Language Generation

Ehud Reiter

(University of Aberdeen)

Springer Cham, 2024, XIII + 202 pp.; Hardcover, ISBN 978-3-031-68581-1; Softcover, ISBN 978-3-031-68584-2; eBook, ISBN 978-3-031-68582-8; doi:10.1007/978-3-031-68582-8

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1. Introduction: Ehud Reiter and Natural Language Generation

This year marks the retirement of Ehud Reiter. As a parting gift, he has written up his thoughts on Natural Language Generation based on 35 years of experience in both research and industry (Reiter 2025a). For Reiter, *Natural Language Generation* (NLG) means: systems that transform data to text, with a clear application context in which that text is to be used. Reiter's practical perspective has strongly influenced the International Natural Language Generation (INLG) conferences, that are held under the banner of ACL's Special Interest Group for Natural Language Generation (SIGGEN). To this day, the conference offers a balanced program with contributions from both academics and industry researchers.

Reiter, as one of the central researchers of this community, has made many important contributions to the field. Together with Robert Dale, he wrote a foundational and often cited book on building applied NLG systems (Reiter and Dale 2000). Moreover, over the course of his long career, he published a number of seminal articles, including for example, Dale and Reiter 1995, on the computational implications of obeying the Gricean maxims when generating referring expressions, and Reiter, Robertson, and Osman 2003, on the development and evaluation of an automatic smoking cessation intervention (the first and perhaps still only NLG application evaluated in a clinical trial). In recent years, he also increasingly wrote about improving evaluating practices and the importance of open and replicable research in NLP. What all these papers have in common is that they are very well-written, in an admirably clear manner. Moreover, they generally do not follow any scientific trends, but rather address topics that Reiter and his co-workers found important to explore at the time.

The book we review here serves both as a personal overview of topics that matter to Reiter and as a timely successor to (Reiter and Dale 2000), which provided a first, general blueprint for the development of NLG technology. As Reiter notes in the Preface, many practical aspects of that book got outdated, but the high-level distinctions introduced by

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Reiter and Dale remain relevant to this day. Rather than trying to describe the current state-of-the-art, this new book aims to provide similarly timeless advice.

The goal for this book is to “present a broad overview of NLG which talks about language models, but also looks at alternative approaches to NLG, requirements (what users want NLG systems to do), evaluation, safety and testing, and *sensible* applications of NLG” (emphasis ours). For Reiter, NLG solutions shouldn’t just look nice on paper, but they should also be useful in practice. Throughout the book, Reiter provides examples from his own research, paired with advice based on his experience in industry. As a result, the book has a somewhat conversational style, in which the author is clearly present, writing from a first person perspective and frequently adding personal notes.

2. First chapters: the roots of NLG

Chapter 1. The first chapter offers a broad overview of the field. In Reiter’s view, Natural Language Generation does not just concern itself with the technicalities of generating text (using insights from AI, NLP, and Software Engineering); NLG researchers must also think about interactions with users (using insights from Human Computer Interaction), consider how the system uses language and how its messages are interpreted (using insights from Linguistics and Psychology), and talk to domain experts to deeply understand the context in which the system is used. (This is why Reiter recently (2025b) argued in this journal that we should also evaluate real-world impact of NLP systems.)

Next to a general introduction to the book, Chapter 1 also presents a brief history of the field. To our knowledge, this is one of the few written accounts of the history of NLG, aside from Robert Dale’s ongoing efforts to record the commercial landscape (e.g., Dale 2020, 2022, 2023).¹ Reiter presents the history of NLG in three phases:

1. The early phase (1960-1990) where the basic rule-based paradigm was being developed.
2. The establishment of NLG (1990-2014) where data-driven methods gained popularity and commercial NLG companies were founded.
3. The current phase after the introduction of neural NLG (2015-now).

Chapter 2. This chapter discusses rule-based NLG systems and makes a strong case for why these are still relevant. In terms of practical benefits, they offer complete control over the output, are testable and auditable, and will not ‘hallucinate.’ Since NLG research (from the third phase onwards) has mostly focused on Machine Learning and neural techniques, we have not seen many developments with respect to rule-based NLG. But rumors of the death of these kinds of systems have been greatly exaggerated: commercial NLG companies, for example, still use them because they offer a level of precision that neural systems are yet to offer. Also in contexts where safety is crucial (such as the health domain) rule-based systems are still a serious option.

Next to the practical benefits of rule-based NLG systems, Reiter also argues that the rule-based approach offers a *conceptual framework* for anyone wanting to build a natural language generation system, regardless of the approach. Either implicitly or explicitly,

¹ In 2021, Reiter also gave an online lecture on the topic that is still available on YouTube: https://www.youtube.com/watch?v=SEw47Y_ZN8Q

every data-to-text system carries out signal analysis, data interpretation, document planning, microplanning, and surface realisation.

3. Machine learning and neural NLG

Most modern NLG systems use at least some form of Machine Learning (ML). The problem with summarising the use of ML in NLG is that any description of ‘current’ technology will soon be outdated. So instead, Reiter only provides a high-level overview and leaves a detailed discussion of ML to other work.² Because of this, the book is able to cover the topic in a single 20-page chapter.

Chapter 3. This chapter starts by introducing three different kinds of tasks: 1. determining what indefinite article (*a* or *an*) an NLG system should use; 2. answering questions about the weather; and 3. generating weather forecasts. These tasks allow Reiter to cover different kinds of ML algorithms that may be employed to carry out NLG tasks, namely: *classifiers*, *N-gram language models*, *early neural models*, and *transformers* (followed by a discussion of foundation models, instruction tuning, and Reinforcement Learning from Human Feedback; RLHF). What matters to Reiter is not how these algorithms work, exactly, but how they can be used to generate texts that fulfill the task requirements. To this end, the book discusses how different ML algorithms can be combined to replace one or more parts of the traditional NLG pipeline (that readers have learned about in Chapter 2), or even the full pipeline (i.e., taking an end-to-end approach).

Machine learning algorithms cannot function without proper training data and, as Reiter tells us, they cannot function *well* without *good* training data. Readers of Reiter’s blog³ will know that poor training data is a frequent source of frustration for him: he strongly argues that data should be of high quality, representative, and there needs to be enough to train a model to perform well. Much of this part of the book *should* be common sense, and indeed it is hard to summarise Reiter’s recommendations without them feeling obvious, but all too often in NLP research data quality is sacrificed at the altar of speed.

At the end of Chapter 3, Reiter discusses issues that may arise when using Machine Learning and neural systems. These issues are especially pertinent for those working with real-life applications. For example, domain shift is a well-known phenomenon in Machine Learning where models may become outdated and performance degrades over time. But more time in NLP research is spent on the development of new models, than on the maintenance of those models. Building on Reiter’s long and fruitful career, the book also offers illustrative anecdotes of issues that may arise in real-world cases. For example, Reiter highlights how even correct texts can be problematic in a medical context: if a system tells patients that ‘ear pain can be a sign of brain tumour or stroke,’ this is factually correct, but doctors will tell you to avoid discussing such low-probability outcomes because this may cause unnecessary stress.

4. Applications

The bulk of the book is focused on applications, starting with user requirements (Chapter 4), and moving through Evaluation (Chapter 5); Safety, testing, and maintenance

² See, for example, Jurafsky and Martin 2026 for a more detailed discussion of ML in NLP.

³ The blog can be found at: <https://ehudreiter.com>

(Chapter 6); with a final chapter on applications themselves (Chapter 7). This order is not accidental: in Reiter's view, good system evaluations are case-specific and informed by user requirements. Once we know whether the system lives up to the requirements, we can further examine the safety of the system. While NLP evaluation is mostly focused on *average-case performance*, software testing expands on this, also looking at *worst-case performance* and thinking about ways to address these cases. Maintenance is the final step, thinking about the upkeep of the model once it has been deployed. The final chapter allows us to reflect on the entire process from start to finish. The value of these chapters lies in the fact that these topics (and particularly chapters 4, 6, and 7) are rarely discussed in the NLP literature, and so Reiter's work is full of insights that expand our vision.

Chapter 4. This chapter provides an overview of the different aspects we need to take into account when developing an NLG system that will actually be used. Reiter starts off by listing common quality criteria for the output of the system (e.g. *readability, accuracy*) as well as the interface (e.g. *responsiveness, operation costs*). He argues that NLG systems should fit into the overall workflow of their users (which also influences the relative importance of these criteria). Since no two use cases are alike, Reiter points out that we need to obtain specific requirements, for which the book offers three directions: carrying out user studies, performing a corpus analysis, and talking to stakeholders. This is not a direct recipe for success, but not talking to different stakeholder groups is a recipe for failure.

Like with some other application-oriented fields (e.g. Human-Computer Interaction), one might stop here and wonder *what is scientific about this?* If NLG is all about developing optimal solutions for specific use cases, how does this lead to generalisable knowledge? Although the book stays away from these kinds of academic questions, we might refer here to the literature on *intermediate-level knowledge*, defined by Höök and Löwgren (2012) as "knowledge that is more abstracted than particular instances, without aspiring to be at the scope of generalized theories."⁴ We can use the examples discussed by Reiter to better understand the design space for NLG systems and how we might approach our own use cases.

Chapter 5. This is by far the largest chapter, being over twice the size as most of the other chapters. This is not surprising: the author has written extensively on evaluation, and counts as one of the leading authorities on this subject. In this journal alone, Reiter has written multiple contributions on automatic evaluation (Reiter and Belz 2009; Reiter 2018), human evaluation (Thomson, Reiter, and Belz 2024), and evaluation in general (Reiter 2025b). Suffice it to say that this chapter provides detailed advice on how to evaluate NLG systems, with plenty of examples and personal notes. The chapter ends with notes on commercial evaluation (which is rarely reported anywhere in the NLG literature) and 'ten tips on evaluating NLG.' In his typical style, the author laments: "I hope that these tips make sense to readers of this book; unfortunately I see *many* published NLG evaluations which do not follow them."

Chapter 6. This chapter is short but sweet. Reiter examines issues related to Safety, Testing, and Maintenance, relying heavily on his experience with commercial NLG systems. In a way, it feels like NLG as a subfield has been ahead of the curve. Since

⁴ Also see the discussion by Dalsgaard and Dindler (2014).

the rise of neural systems, and particularly since the rise of ChatGPT, the entire field of Natural Language Processing has needed to come to terms with the fact that NLP has real-world impact now.⁵ But (as we have learned in Chapter 1) commercial NLG systems have been around for a bit longer, meaning that they were already working on issues like safety. NLG companies are careful to adopt new technology, exactly because of the reasons discussed in this chapter: systems need to be reliable, and it should not be overly burdensome to support and maintain actively deployed systems. If a client asks for changes to be made to the system, it should be relatively straightforward to do so. Fully retraining models to incorporate changes may not be feasible.

Chapter 7. Having read the previous six chapters, readers are rewarded with high-level reflections on practical applications of NLG. Reiter first discusses general challenges in building a successful NLG system, before he moves to practical applications in Journalism, Business Intelligence, Summarisation, and Healthcare (the latter is still an underexplored application domain of NLG, despite the clear promise). This part is where Reiter's experience really pays off, since he is able to draw from a wide range of applications and provide anecdotes showing how to not only get NLG systems to work but also how to get people to actually use them. We might illustrate this with two anecdotes:

“Personal Note: Many years ago, the Scottish AI pioneer Rob Milne (now sadly deceased) told me that the most successful AI applications were those that targeted ‘peripheral’ tasks, i.e. tasks that users saw as annoying distractions. It was much harder to build a successful AI application for a core task, i.e. a task that the user saw as a central aspect of his job and perhaps identity. Diagnosis is a core task for doctors.”

“Personal Note: Patients may want to see some medical terminology, even if they do not fully understand it. For example, in our Babytalk system for parents [...], we initially tried to remove medical terminology. However, some parents didn't like this and said that we were ‘dumbing down’ texts and patronising them, so we put some medical terminology back into the parent reports, even though in some cases this made them harder to understand.”

5. Conclusion

Overall, this book is engaging and well written. Reiter takes a clear no-nonsense approach, combining his general expertise with personal reflections on the field. Readers of his blog will find the style to be very recognizable. Since the book is strongly geared towards applications, it does not discuss more theoretical sub-fields of Natural Language Generation, such as cognitive modeling. The book does not talk about the similarities and differences between human and automatic language production, and how one might make systems produce more human-like texts. Instead, it appeals to a more urgent need: with the outsized impact of Large Language Models on NLP, we have seen many newcomers to the field who are (seemingly) unaware of its history.⁶ This book provides a solid starting point and a warm welcome for readers interested in the tradition and the potential of the field to produce useful systems.⁷

⁵ The work by Hovy and Spruit (2016) on the Social Impact of NLP is a clear sign of the times.

⁶ This situation has also prompted different studies into ‘citation amnesia’ in NLP, see e.g. Bollmann and Elliott 2020; Wahle et al. 2025.

⁷ Contributing to this welcoming, the book is also available as a preprint via ArXiv.

It is perhaps ironic that after working for decades on applied, often rule-based text generation systems (even when the rest of NLP was already moving towards statistical, machine learning and eventually neural systems), we have now entered an era in which text generation is perhaps the best known and most studied NLP application. And even though the methods used by ChatGPT and its ilk are a far cry from the methods developed and studied by Reiter, his insights remain as relevant as ever: we still need to think about real world applications, good scientific practice (open, replicable, paying attention to data quality) continues to be paramount, and proper multi-method evaluation approaches remain key to progress.

Acknowledgments

Thanks to the reading group on the SIGGEN Discord server for discussion.

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